No
Date

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Mata kuliah : Teknik Pengukuran Frekuensi Tinggi

Tugas 2

Melakukan analisa dan menghitung kembali contoh pembahasan untuk problem 2a dan 2b berikut ini.

2A. Dik : A = Directivity = 30 dB $\approx 0{,}032$

$C = \rho_x \approx \sqrt{\sigma r^2 + (0{,}75 . \rho_c)^2}$

$0{,}10 = \sqrt{\sigma r^{\circ} + (0{,}75 \rho_c)^2}$

$\rho_c = 0{,}126$

$B = A + C = 0{,}032 + 0{,}10$

$= 0{,}132$

$\Delta\rho = A + B\rho_L + C . \rho_L^2$

$\Delta\rho = 0{,}032 + 0{,}132\rho_L + 0{,}10\rho_L^2$

$\rightarrow \rho_L = 0{,}5$

* tanpa instrument error (B) → best condition

$\Delta\rho = A + C\rho_L^2$

$= 0{,}032 + 0{,}10 . (0{,}5)^2$

$= \pm 0{,}057$

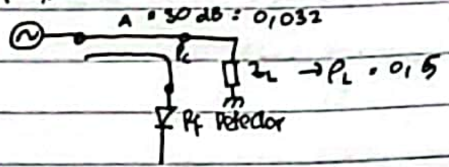
* dengan instrument error (B) → worst case condition

$\Delta\rho = A + B\rho_L + C\rho_L^2$

$= 0{,}032 + 0{,}132 (0{,}5) + 0{,}10 (0{,}5)^2$

$= \pm 0{,}123$

* pengukuran akurasi suatu beban

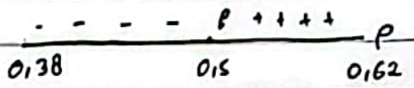


$\rho_c = 0{,}13$ → yg ada pada Directional coupler

A: Directivity
B = error Instrument kalibrasi
C = effective source match

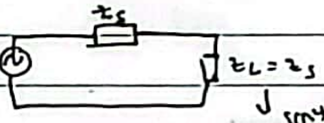
$\Delta\rho$: worst case condition $\overset{\text{short}}{\text{cal}}$ / open cal
↳ error dari kalibrasi bisa diabaikan klo kalibrasinya barengan antara short dan open.

error window ρ



* perhitungan ρ paling besar hanya 2 digit dibelakang koma

*



↓
~~semua akan diserap oleh beban~~ sinyal

$Z_L \neq Z_s$ (~~sinyal~~ sebagian sinyal akan dikembalikan kesumber)

2B. Dik : A = Directivity = 36 dB = 0,016
output SWR = 1,5
$\rho_L = 1{,}9$

- $\rho_x = C = \sqrt{\sigma r^2 + (0{,}75\rho_0)^2}$
$= 0{,}15$

$\rho = \dfrac{Z_L - Z_s}{Z_L + Z_s}$

$V_{SWR} = \dfrac{1 + \rho}{1 - \rho}$

$|\rho| = \dfrac{V_{SWR} - 1}{V_{SWR} + 1}$

$0 < \rho < 1$

$RL = -20 . \log |\rho|$

JOYKO 36 Lines, 6 mm

$P_x \cdot \sqrt{(0{,}15)^2 + (0{,}75 P_c)^2}$

$0{,}15 \cdot \sqrt{(0{,}016)^2 + (0{,}75 P_c)^2}$

$P_c = 0{,}199$

$\cdot B = A + C = 0{,}016 + 0{,}15$

$= 0{,}166$

maka ΔP

$\Delta P = A + B P_L + C P_L^2$

$= 0{,}016 + 0{,}166 P_L + 0{,}15 P_L^2$

$\rightarrow P_L = 1{,}9$ (VSWRnya)

$P_L = 0{,}31$

* tanpa instrumen error (B)

$\Delta P = A + C P_L^2$

$= 0{,}016 + 0{,}15 (1{,}9)^2$

$= \pm\ 0{,}558$

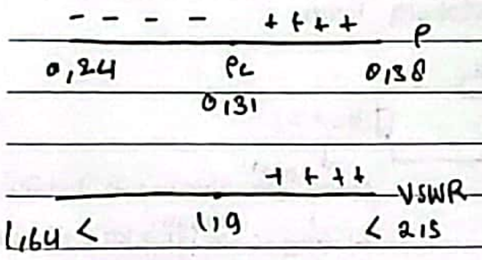
* dengan instromen error (B)

$\Delta P = A + B P_L + C P_L^2$

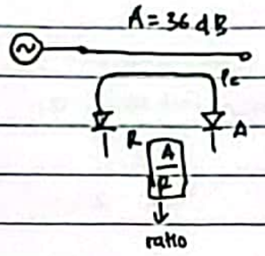
$= 0{,}016 + 0{,}166 (1{,}9) + 0{,}15 \cdot (1{,}9)^2$

$= \pm\ 0{,}873$

error window



* Dual coupler → Ar ratkon



$P_c = 0{,}19$ dB

* km menggunakan ratio measurement maka menggunakan kalibrasi short dan open

↳ hasil karibrasi merupakan return loss

* B gk usah dihitung

$\Delta P = A + C P_L^2$

$= 0{,}01 + 0{,}15 (0{,}31)^2$

$= 0{,}024$ (total uncertainty)

* B dihitung

$\Delta P = A + B P_L + C P_L^2$

$= 0{,}01 + 0{,}16 (0{,}31) + 0{,}15 (0{,}31)$

$= 0{,}074$

CS CamScanner